



CODECHEF

The CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest, is now in print!

*H*ello and welcome to the *CodeChef Puzzle Challenge* sponsored by *LINUX For You* and *Directi*! Solve this month's puzzle, and you could be one of the three lucky people to win a cash prize of Rs 1,000 each!

The May edition's puzzle:

A man is standing on the bank of a river. He sees a temple directly in front of him on the opposite bank. He wants to visit the temple. He decides to row a boat to reach the other bank. He can row at a speed of 4m/sec and the river water flows at a speed of 3m/sec. The man's walking speed is 2m/sec. He starts rowing the boat in the straight direction towards the temple.

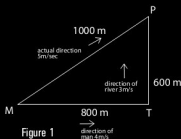
Given that the width of the river is 800 metres, what is the time taken by the man to reach the temple?

The solution:

Refer Figure 1.

Let M be the man and T be the temple on the opposite bank of the river. The man wants to reach T but the river is flowing at a speed of 3m/s.

Due to the flow of the river, the man reaches point P (i.e., he drifts slightly off-course). Now the time taken by the man to cross the river can be



calculated by considering the direction MT only (using vectors) = $MT/4 = 800/4 = 200 \text{ sec} \dots (1)$

In 200 sec, the man drifts to point P, so the distance $PT = 3 \times 200 = 600 \text{ m} \dots (2)$

Now the man walks at a speed of 2 m/s, so the time taken by him to travel from P to T = $PT/2 = 600/2 = 300 \text{ sec} \dots (3)$

So the total time required to reach T = eqn 1 + eqn 3 = $200 + 300 = 500 \text{ seconds}$.

And the winners this month are:

- Anurag Srivastava
- Ruhani Arora
- Jagannath Prasad Das

Here's Your CodeChef Puzzle of the Month:

A boy lives on the third floor of a building at a height of 12 metres from the ground. He spots a hot air balloon in the sky at an inclination of 45° . At the same time, a man standing at the bottom of the building spots it at an inclination of 60° . The boy runs inside to his mother to tell her about what he's seen. After 10 seconds, his mother comes out to have a look at the balloon. The angle of inclination of the balloon remains the same, i.e., 45° . Can you find the altitude of the balloon from the ground when the mother sees it? Consider the horizontal speed of the balloon to be 3 m/sec. Also, find the vertical speed of the balloon.

Looking for a little extra cash?

Solve this month's *CodeChef Challenge* puzzle and send in your answer to codechef@efyindia.com or lly@codechef.com by June 15, 2012. Three lucky winners will get some awesome prizes both in cash and merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou>. **END** 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the June Challenge that takes place from the 1st to the 11th of every month to win cash prizes up to Rs 20,000. Also keep visiting CodeChef and participate in the multiple programming contests that take place on the website throughout the month!